1.Singleton.

Student.java | TasksingletonApplication.java

```java
package com.example.Tasksingleton;
import org.springframework.context.annotation.Scope;
import org.springframework.stereotype.Component;
@Component
@Scope("singleton")
public class Student {
    int Student_id;  4 usages
    int roll_no;  4 usages
    String student_name;  4 usages
    String student_department;  4 usages

    public void display_details() {  2 usages
        System.out.println();
        System.out.println("Student id: " + Student_id);
        System.out.println("Roll no: " + roll_no);
        System.out.println("Student name: " + student_name);
        System.out.println("Department: " + student_department);

    }

}
```

```java
package com.example.Tasksingleton;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;

@SpringBootApplication
public class TasksingletonApplication {

    public static void main(String[] args) {

        ApplicationContext context =
                SpringApplication.run(TasksingletonApplication.class, args);

        Student s = context.getBean(Student.class);

        s.Student_id = 1001;
        s.roll_no = 220002;
        s.student_name = "SaminSathik";
        s.student_department = "BCA";

        Student s1 = context.getBean(Student.class);
```

```
        Student s = context.getBean(Student.class);

        s.Student_id = 1001;
        s.roll_no = 220002;
        s.student_name = "SaminSathik";
        s.student_department = "BCA";

        Student s1 = context.getBean(Student.class);

        s1.Student_id = 1002;
        s1.roll_no = 220003;
        s1.student_name = "Kuthubudeen";
        s1.student_department = "IT";

        s1.display_details();
        s.display_details();
    }
}
```

TasksingletonApplication ×

output:

```
Student id: 1002
Roll no: 220003
Student name: Kuthubudeen
Department: IT
```

2.Prototype

Task.java × TaskPrototypeApplication.java

```java
package com.example.TaskPrototype;
import org.springframework.context.annotation.Scope;
import org.springframework.stereotype.Component;
@Component
@Scope("prototype")
public class Task {

    public int task_id;  3 usages
    public String task_name;  3 usages
    public String assigned_to;  3 usages
    public String task_status;  3 usages

    public void display_details() {  2 usages

        System.out.println("Task ID: " + task_id);
        System.out.println("Task Name: " + task_name);
        System.out.println("Assigned To: " + assigned_to);
        System.out.println("Task Status: " + task_status);

        System.out.println("-------------------------");
    }
}
```

Task.java | TaskPrototypeApplication.java

```java
package com.example.TaskPrototype;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;

@SpringBootApplication
public class TaskPrototypeApplication {

    public static void main(String[] args) {

        ApplicationContext context =
                SpringApplication.run(TaskPrototypeApplication.class, args);

        Task t = context.getBean(Task.class);

        t.task_id = 101;
        t.task_name = "Create Login Page";
        t.assigned_to = "Samin";
        t.task_status = "Completed";

        Task t1 = context.getBean(Task.class);
```

```
        Task t = context.getBean(Task.class);

        t.task_id = 101;
        t.task_name = "Create Login Page";
        t.assigned_to = "Samin";
        t.task_status = "Completed";

        Task t1 = context.getBean(Task.class);

        t1.task_id = 102;
        t1.task_name = "Create Registration Page";
        t1.assigned_to = "Kuthubudeen";
        t1.task_status = "Pending";

        t.display_details();
        t1.display_details();
    }
}
```

Output:

```
2026-10-21T20:40:49.521+05:30  INFO 4424 --
Task ID: 101
Task Name: Create Login Page
Assigned To: Samin
Task Status: Completed
-------------------------
Task ID: 102
Task Name: Create Registration Page
Assigned To: Kuthubudeen
Task Status: Pending
-------------------------
```

3.Component:

BankAccount.java × BankAccountManagementApplication.java application.prop

```java
package com.example.BankAccountManagement;

import org.springframework.stereotype.Component;

@Component
public class BankAccount {

    public int account_number;  2 usages
    public String account_holder;  2 usages
    public String account_type;  2 usages
    public double account_balance;  2 usages

    public void display_details() {  1 usage

        System.out.println("Account Number: " + account_number);
        System.out.println("Account Holder: " + account_holder);
        System.out.println("Account Type: " + account_type);
        System.out.println("Account Balance: " + account_balance);
    }
}
```

BankAccount.java | BankAccountManagementApplication.java | application.properties

```java
package com.example.BankAccountManagement;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;

@SpringBootApplication
public class BankAccountManagementApplication {

    public static void main(String[] args) {

        ApplicationContext context =
                SpringApplication.run(BankAccountManagementApplication.class, args);

        BankAccount b = context.getBean(BankAccount.class);

        b.account_number = 10001;
        b.account_holder = "Samin";
        b.account_type = "Savings";
        b.account_balance = 25000.50;

        b.display_details();
    }
}
```

Output:

```
Account Number: 10001
Account Holder: Samin
Account Type: Savings
Account Balance: 25000.5
```

THEORY QUESTIONS

1.What is Spring Boot?

Spring Boot is a Java framework used to develop applications quickly. Advantages include auto-configuration, embedded servers, less code and easy setup.

2.Spring Framework and Spring Boot?

Framework requires more manual configuration, whereas Spring Boot provides auto-configuration and embedded servers.

3.@SpringBootApplication?

It is the main annotation that combines @Configuration, @EnableAutoConfiguration and @ComponentScan.

4.(DI) and (IoC)?

IoC means Spring manages objects, while DI means Spring provides the required dependencies to objects.

5.@Component, @Service and @Repository?

@Component is a general bean annotation, @Service is used for business logic and @Repository is used for database operations

6.@Controller and @RestController?

@Controller is mainly used for web pages, whereas @RestController is used to return data such as JSON directly.